Phase 3, Block 3: Crank–Nicolson, the θ-scheme, and Rannacher time-stepping

Quant Finance Portfolio

1 Setting and motivation

Blocks 1 (FTCS) and 2 (BTCS) are first-order in time: their local truncation error is $O(\Delta\tau) + O(\Delta x^2)$. To balance the spatial and temporal contributions one needs $\Delta\tau \sim \Delta x^2$, which forces $M \sim N^2$ and total cost $O(N^3)$ for both schemes (FTCS by CFL, BTCS by accuracy).

This block implements *Crank–Nicolson* (CN), the unique scheme in the convex family of FTCS and BTCS that is *second-order in time*: $O(\Delta\tau^2) + O(\Delta x^2)$. With $M \sim N$, CN achieves $O(\Delta x^2)$ precision at $O(N^2)$ total cost — a factor-N improvement over FTCS/BTCS in the same precision regime. CN is the workhorse of finite-difference option pricers in production.

The deliverable is a θ-scheme infrastructure (`quantlib.theta_scheme`) that subsumes FTCS, BTCS and CN as parameter choices, plus the CN pricer with Rannacher time-stepping (`quantlib.cn`) to suppress spurious oscillations near the strike kink.

2 The θ-family

The three schemes of Phase 3 are particular cases of a single parametric family:

$$\frac{V_j^{n+1} - V_j^n}{\Delta\tau} \;=\; \theta\,\mathcal{L}_h V_j^{n+1} \;+\; (1-\theta)\,\mathcal{L}_h V_j^n, \qquad \theta \in [0,1], \tag{1}$$

where $\mathcal{L}_h$ is the discrete spatial operator

$$\mathcal{L}_h V_j \;=\; \frac{\sigma^2}{2}\,\frac{V_{j+1} - 2V_j + V_{j-1}}{\Delta x^2} + \mu\,\frac{V_{j+1} - V_{j-1}}{2\Delta x} - r\,V_j.$$

The cases of interest are

θ	Name	Type	Stability	Time order
0	FTCS	Explicit	Conditional ($\alpha \le 1/2$)	$O(\Delta\tau)$
1/2	**Crank–Nicolson**	Implicit	Unconditional	$O(\Delta\tau^2)$
1	BTCS	Implicit	Unconditional	$O(\Delta\tau)$

The order jump at $\theta = 1/2$ is the central observation of this block. It is the pay-off of the symmetry between the new and old levels: when the right-hand side is the average of $\mathcal{L}_h V$ at τ_n and τ_{n+1}, the discrete equation is centred at the *midpoint* $\tau_{n+1/2}$, and the midpoint rule is one order more accurate than the endpoint rules.

3 Crank–Nicolson explicitly

Setting $\theta = 1/2$ in (1), expanding $\mathcal{L}_h$, and rearranging V^{n+1} to the left, V^n to the right:

$$\boxed{\beta_-\,V_{j-1}^{n+1} + \beta_0\,V_j^{n+1} + \beta_+\,V_{j+1}^{n+1} \;=\; \gamma_-\,V_{j-1}^n + \gamma_0\,V_j^n + \gamma_+\,V_{j+1}^n.} \tag{2}$$

With $\alpha = \frac{\sigma^2}{2}\,\Delta\tau/\Delta x^2$ and $\nu = \mu\,\Delta\tau/(2\Delta x)$ as before, the LHS coefficients are

$$\beta_- = -\tfrac{1}{2}\alpha + \tfrac{1}{2}\nu, \quad \beta_0 = 1 + \alpha + \tfrac{1}{2}r\Delta\tau, \quad \beta_+ = -\tfrac{1}{2}\alpha - \tfrac{1}{2}\nu,$$

and the RHS coefficients are

$$\gamma_- = +\tfrac{1}{2}\alpha - \tfrac{1}{2}\nu, \quad \gamma_0 = 1 - \alpha - \tfrac{1}{2}r\Delta\tau, \quad \gamma_+ = +\tfrac{1}{2}\alpha + \tfrac{1}{2}\nu.$$

Symmetry: $\beta_\pm = -\frac{1}{2}$(the corresponding BTCS coefficient minus 1 on diagonal), $\gamma_\pm = +\frac{1}{2}$(the corresponding F
CN is, structurally, "half BTCS plus half FTCS".

4 Why CN is second-order in time: the midpoint rule

The non-trivial claim is that the local truncation error of (2) is $O(\Delta\tau^2) + O(\Delta x^2)$, despite the fact that the time difference $(V^{n+1} - V^n)/\Delta\tau$ looks like the same first-order forward Euler quotient that FTCS uses. The gain comes from *which time level the equation is centred on.*

Expand V around the midpoint $\tau_{n+1/2} = \tau_n + \Delta\tau/2$:

$$\begin{aligned} V(\tau_{n+1}) &= V_* + \tfrac{1}{2}\Delta\tau\, V_*' + \tfrac{1}{8}\Delta\tau^2\, V_*'' + \tfrac{1}{48}\Delta\tau^3\, V_*''' + O(\Delta\tau^4), \\ V(\tau_n) &= V_* - \tfrac{1}{2}\Delta\tau\, V_*' + \tfrac{1}{8}\Delta\tau^2\, V_*'' - \tfrac{1}{48}\Delta\tau^3\, V_*''' + O(\Delta\tau^4), \end{aligned}$$

where $V_* := V(\tau_{n+1/2})$, $V_*' := V_\tau(\tau_{n+1/2})$, etc. Subtracting and dividing by $\Delta\tau$:

$$\frac{V(\tau_{n+1}) - V(\tau_n)}{\Delta\tau} \;=\; V_\tau(\tau_{n+1/2}) \;+\; \frac{\Delta\tau^2}{24}\, V_{\tau\tau\tau}(\tau_{n+1/2}) \;+\; O(\Delta\tau^4). \tag{3}$$

The $\Delta\tau$ term *cancels by symmetry*: the quotient is a second-order approximation to V_τ at the midpoint.

Analogously, adding the two expansions of $\mathcal{L}_h V$ (one at τ_n, one at τ_{n+1}) and dividing by 2:

$$\tfrac{1}{2}[\mathcal{L}_h V(\tau_{n+1}) + \mathcal{L}_h V(\tau_n)] \;=\; \mathcal{L}_h V(\tau_{n+1/2}) \;+\; O(\Delta\tau^2) \;+\; O(\Delta x^2), \tag{4}$$

where the $O(\Delta\tau^2)$ comes from the time interpolation (linear average of the endpoints) and the $O(\Delta x^2)$ is the spatial truncation of $\mathcal{L}_h$. Combining (3) and (4), the CN equation $(V^{n+1} - V^n)/\Delta\tau = \frac{1}{2}(\mathcal{L}_h V^{n+1} + \mathcal{L}_h V^n)$ is the discretisation of the exact PDE $V_\tau = \mathcal{L}V$ *at the midpoint*, with truncation

$$\tau_h = O(\Delta\tau^2) + O(\Delta x^2).$$

FTCS and BTCS, by contrast, evaluate the right-hand side at τ_n or τ_{n+1} (not the midpoint); the time error there is $O(\Delta\tau)$.

5 The linear system and matrix structure

Equation (2) written for the interior nodes $j = 1, \ldots, N-1$ becomes

$$A\,\mathbf{V}_{\text{int}}^{n+1} \;=\; B\,\mathbf{V}_{\text{int}}^{n} + \mathbf{b}_{\text{bnd}}^{n},$$

where A and B are tridiagonal $(N-1)\times(N-1)$ matrices with $(\beta_-, \beta_0, \beta_+)$ and $(\gamma_-, \gamma_0, \gamma_+)$ on their diagonals, and the boundary vector mixes both time levels:

$$\mathbf{b}_{\text{bnd}}^{n} = \begin{pmatrix} \gamma_-\, V_0^n - \beta_-\, V_0^{n+1} \\ 0 \\ \vdots \\ 0 \\ \gamma_+\, V_N^n - \beta_+\, V_N^{n+1} \end{pmatrix}. \tag{5}$$

Both lateral coefficients of the stencils contribute to the boundary correction, in contrast to BTCS where only the LHS coefficients appeared.

The per-step algorithm has three phases:

1. Apply the RHS stencil B to $\mathbf{V}^n_{\text{int}}$ as a three-point stencil sweep ($O(N)$, no need to assemble B explicitly).
2. Add the boundary correction $\mathbf{b}^n_{\text{bnd}}$ ($O(1)$: only two endpoints are non-zero).
3. Solve $A\,\mathbf{V}^{n+1}_{\text{int}} =$ (result) by Thomas with the pre-factored A ($O(N)$).

Total cost per step: $O(N)$. Total cost: $O(N \cdot M)$, the same as BTCS.

Pre-factoring A. The matrix A is constant in time (again thanks to the constant-coefficient form of the transformed PDE). Factoring A once at setup and applying the factor at every step is the standard pattern, identical to BTCS.

6 Stability of the θ-scheme

The von Neumann amplification factor of (1), ignoring the reactive term, is

$$g(\xi) = \frac{1 - 4\alpha(1-\theta)\sin^2(\xi/2) + i\,(1-\theta)\beta_C \sin\xi}{1 + 4\alpha\theta\sin^2(\xi/2) - i\,\theta\beta_C \sin\xi}, \tag{6}$$

with $\beta_C := \mu\Delta\tau/\Delta x$, $\xi \in [-\pi, \pi]$ the Fourier wavenumber.

Proposition 6.1 (Stability of the θ-scheme)**.** *The θ-scheme is unconditionally von Neumann stable if and only if $\theta \geq 1/2$. For $\theta < 1/2$, $|g| \leq 1$ requires the conditional bound*

$$\alpha \;\leq\; \frac{1}{2(1-2\theta)}.$$

In particular, $\theta = 0$ gives $\alpha \leq 1/2$ (the FTCS CFL condition), and $\theta = 1/2$ is the smallest θ for which the bound is vacuous.

Sketch. Set $s = \sin^2(\xi/2)$, $u = \sin\xi$, so $u^2 = 4s(1-s)$. Then $|g|^2 = N(s,u)/D(s,u)$ where both numerator and denominator are quadratic in α. The condition $|g|^2 \leq 1$ becomes $D - N \geq 0$, which after expansion is

$$8\alpha s\left[(2\theta - 1) + 4\alpha\theta(1-\theta)s\right] \;+\; (\text{terms vanishing when } \beta_C = 0) \;\geq\; 0.$$

For $\theta \geq 1/2$, both bracketed terms are non-negative for any $\alpha \geq 0$, $s \in [0,1]$. For $\theta < 1/2$, the first term is negative and the bound becomes the stated inequality. (Full algebra in Strikwerda §6.3, or [4, §4.5].) □ □

CN at $\theta = 1/2$ is therefore *exactly on the boundary* between conditional and unconditional stability. This is the structural reason why CN, although stable, has weaker damping properties than BTCS, as we now examine.

7 The kink-induced oscillation problem

The amplification factor at the Nyquist wavenumber $\xi = \pi$ (highest representable frequency on the grid, $\sin^2(\pi/2) = 1$, $\sin\pi = 0$) is

$$g_\theta(\pi) = \frac{1 - 4\alpha(1-\theta)}{1 + 4\alpha\theta}.$$

For BTCS ($\theta = 1$) and CN ($\theta = 1/2$):

$$g_{\text{BTCS}}(\pi) = \frac{1}{1+4\alpha} \xrightarrow{\alpha\to\infty} 0, \qquad g_{\text{CN}}(\pi) = \frac{1-2\alpha}{1+2\alpha} \xrightarrow{\alpha\to\infty} -1.$$

BTCS damps the Nyquist mode exponentially; CN preserves it with a sign flip. In the regime of interest for CN ($\Delta\tau \sim \Delta x$, so $\alpha \sim 1/\Delta x \to \infty$ as $N \to \infty$), $|g_{\mathrm{CN}}(\pi)|$ is essentially 1, and the Nyquist mode is preserved indefinitely while alternating sign in time.

This is harmless for smooth initial data: the Fourier coefficient of the Nyquist mode in the initial vector V^0 is exponentially small, and "preserving exponentially small" yields nothing visible.

The payoff is not smooth. A vanilla call payoff $\phi(x) = K(e^x - 1)^+$ has a kink at $x = 0$: ϕ is continuous but ϕ' has a jump. The Fourier expansion of a function with such a kink has coefficients decaying only as $1/k^2$, not exponentially. The Nyquist coefficient in V^0 is therefore $O(1/N^2)$ — not exponentially small.

Consequence. An $O(1/N^2)$ amplitude at the Nyquist mode, preserved across M time steps with sign flip $g \approx -1$ at each, produces a visible oscillation pattern in the computed V^M:

- Bounded ($|g| \leq 1$, so the oscillation does not grow).
- Persistent (each step preserves the magnitude).
- Localised near the kink (where the kink-induced mode lives).
- Alternating sign at consecutive grid nodes (the spatial signature of $\xi = \pi$).

The visible artefact is a sawtooth of small amplitude near the strike, distinct from the explosive sawtooth of FTCS instability (Phase 3 numerical lesson 1) by being bounded but visible. It degrades local accuracy near the strike: the price has $O(\Delta x)$ error there instead of $O(\Delta x^2)$, and the Greeks (gamma in particular) inherit the oscillation.

8 Rannacher time-stepping

The standard fix, due to Rannacher [5], is to replace the first *two* CN time steps by *four* half-time-step BTCS steps, each of length $\Delta\tau/2$:

$$\underbrace{\mathrm{BTCS}_{\Delta\tau/2} \circ \mathrm{BTCS}_{\Delta\tau/2} \circ \mathrm{BTCS}_{\Delta\tau/2} \circ \mathrm{BTCS}_{\Delta\tau/2}}_{\text{4 half-steps, total time } 2\Delta\tau} \circ \underbrace{\mathrm{CN}_{\Delta\tau} \circ \cdots \circ \mathrm{CN}_{\Delta\tau}}_{M-2 \text{ full steps}}. \tag{7}$$

Why it works. Each BTCS half-step damps the Nyquist mode by a factor $g_{\mathrm{BTCS},\Delta\tau/2}(\pi) = 1/(1+2\alpha)$ (α is the CN Fourier number; the half-step uses $\alpha/2$, giving $1/(1+2\alpha)$). After four half-steps the Nyquist mode is multiplied by $1/(1+2\alpha)^4$, which for $\alpha = O(N) \gg 1$ is $O(1/N^4)$ — well below the $O(\Delta x^2) = O(1/N^2)$ floor of the discretisation. By the time CN takes over, the kink-induced high-frequency modes are damped to round-off, and CN proceeds with effectively smooth data.

Why two full steps. Giles & Carter [6] prove that this is the minimum number that recovers $O(\Delta\tau^2)$ convergence in L^2 globally. For pointwise L^∞ convergence at the kink itself, four full steps of Rannacher would be needed; L^2 is sufficient for the price at any fixed spot, recovered by interpolation that smooths over the kink anyway.

Why BTCS, not finer CN. The artefact is not a function of $\Delta\tau$ being too large; it is the absence of damping. CN with arbitrarily small $\Delta\tau$ still has $|g_{\mathrm{CN}}(\pi)| = 1$ in the limit (the bound is sharp). Only a dissipative scheme can remove the irregular high-frequency content, and BTCS is the natural choice because the same Thomas factorisation can be reused with the appropriate coefficients.

Cost. Four half-steps of BTCS cost as much as two full CN steps. For $M \geq 10$ the overhead is below 5%; for $M \geq 100$ below 0.5%. Negligible compared to the convergence gain.

9 Implementation outline

The block ships two modules:

- `quantlib.theta_scheme` (Python) / `quant::pde::theta_scheme_*` (C++): a generic θ-stepper parametrised by θ. Internally pre-factors the LHS matrix and runs a configurable number of time steps. Reusable as scaffolding for Block 4 (where the LHS solve is replaced by PSOR).
- `quantlib.cn` (Python) / `quant::pde::cn_*` (C++): the user-facing CN pricer for European options, with Rannacher smoothing enabled by default (`rannacher_steps=2`). Setting `rannacher_steps=0` gives vanilla CN, exposed to make the oscillation phenomenon testable.

The signature mirrors BTCS:

```
cn_european_call(S, K, r, sigma, T, *, N, M, n_sigma=4.0,
                 rannacher_steps=2)
```

The θ-scheme module exposes a low-level kernel `theta_march(grid, V0, theta, bc_lower, bc_upper)` that the CN pricer calls twice for the Rannacher protocol: once with $\theta = 1$ for the initial half-stepped BTCS warm-up, then with $\theta = 1/2$ for the rest.

No replacement of FTCS/BTCS. The FTCS and BTCS modules of Blocks 1 and 2 are unchanged. The θ-scheme infrastructure is a parallel mechanism intended for CN and any future generalisations. The previous granular commits remain valid; this block adds, does not modify.

10 Validation strategy

Six tests certify the new pricer:

1. Cross-validation against Black–Scholes closed form on a parameter grid (the standard battery of Blocks 1–2).
2. **Quadratic convergence in time:** halving $\Delta\tau$ at fixed (small) Δx should reduce the error by a factor of ≈ 4, not ≈ 2 as BTCS would. This is *the* test of CN.
3. **Balanced quadratic convergence:** doubling both N and M should reduce the error by ≈ 4, vs the ≈ 2 expected for BTCS in the same configuration.
4. **Vanilla CN (without Rannacher) has degraded convergence:** running the same problem with `rannacher_steps=0` should show a noticeably worse convergence ratio. This is the empirical evidence of the kink-induced oscillation.
5. Reproduction of BTCS via $\theta = 1$ in the θ-scheme: verifies the generic stepper against the Block 2 pricer.
6. Put-call parity, input validation, etc.

The fourth test is the operational signature of Block 3's numerical lesson, analogous to the CFL-violation diagnostic of Block 1: a configuration that produces visibly degraded output, captured as a passing assertion to prevent regression.

References

References

[1] D. Tavella and C. Randall. *Pricing Financial Instruments: The Finite Difference Method.* Wiley, 2000. §3.3 (CN), §3.5 (the θ-scheme), §6.3 (Rannacher and other smoothing techniques).

[2] R. Seydel. *Tools for Computational Finance.* 6th ed., Springer, 2017. §4.2.3 (CN), §4.5.4 (kink and Rannacher in practice).

[3] J. C. Strikwerda. *Finite Difference Schemes and Partial Differential Equations.* 2nd ed., SIAM, 2004. §6.3 (stability analysis of the θ-method).

[4] G. D. Smith. *Numerical Solution of Partial Differential Equations: Finite Difference Methods.* 3rd ed., OUP, 1985. §1.13 (the θ-family of schemes).

[5] R. Rannacher. *Finite element solution of diffusion problems with irregular data.* Numerische Mathematik 43, 309–327, 1984. The original paper introducing the smoothing protocol.

[6] M. B. Giles and R. Carter. *Convergence analysis of Crank-Nicolson and Rannacher time-marching.* Journal of Computational Finance 9(4), 89–112, 2005. Sharp results for option pricing: proves that two Rannacher steps suffice for L^2-recovery of $O(\Delta\tau^2)$ convergence.